

Informer-based Anomaly Detection for Internet TV Service in Home Network

Guosong Lv¹, Jing Wang¹, Xingyu Wang², Chengsen Wang²,
Xinxin Chen¹, Jiankun Li¹, Dewei Chen¹, Bo Wu¹, LianYuan Li¹, Wei Yu¹

¹China Mobile Research Institute, China

²Beijing University of Posts and Telecommunications, China

¹Email: {lvguosong, wangjing, chenxinxin, lijiankun, chendewei, wub, lilianyuan, yuwei, yf}@chinamobile.com

²Email: {wangxingyu, cswang}@bupt.edu.cn

Abstract—With more and more users watch live and on-demand content through set-top boxes, internet TV has become one of China Mobile's main home internet services. Meanwhile, the problems of video freezing and blurred screen have increasingly dominated the user complaints. Because of the large number of network elements involved in Internet TV service, traditional quality inspection methods which mainly consist of manual operation and semiautomatic procedures failed to find abnormalities quickly and accurately. In this paper, we collect set-top box active users and its alarm events through China Mobile's soft probe monitoring system and accurately detect the quality degradation abnormalities in TV service by an Informer-based model, the precision rate of anomaly detection reaches over 80%, and the accuracy rate reaches over 90%.

Index Terms—anomaly detection, video freezing, blurred screen, informer, TV service, quality degradation

I. INTRODUCTION

With the development in recent years, China Mobile's home internet TV service has transitioned from the scale growth stage to the value operation stage. As a result the service needs to focus more on the content and the stability and fluency of the content broadcast. Therefore, it is necessary to achieve effective monitoring of user experience and service quality stability and degradation of TV service quality, and further find the network elements that may cause quality degradation.

Fig. 1 below shows the overall network architecture of China Mobile's home broadband Internet TV service. From the figure, It can be seen that the quality of the home broadband Internet TV service involves many network elements, including terminal devices, transmission network devices, CDN devices and service platform servers. The network operation and quality inspection need to deal with the characteristics of a long process, multi-layer and complex network element connection. In addition, China Mobile's home Internet TV service is deployed and maintained on a provincial basis, the number of subscribers in each province varies, and the number of TV service quality issues will change over time and vary among provinces, which make traditional manual operation and testing method impractical in meeting the real-time demands of anomaly detection. Therefore, intelligent methods are urgently needed to monitor and analyze TV service quality. The ultimate goal of the intelligent algorithm is to quickly detect the serious degradation anomaly and provide

timely feedback to operations and maintenance staff so as to troubleshoot and resolve the problems within several minutes.

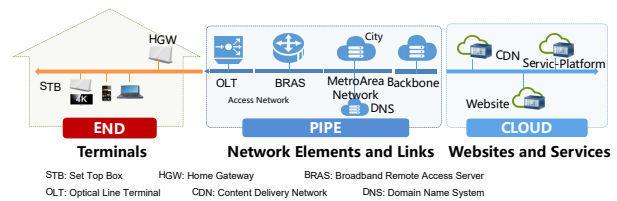


Fig. 1. Network architecture of Internet TV service

To achieve effective detection of the quality of home broadband TV services, we developed a soft probe system based on set-top boxes and home gateways. The goal is to use the set-top box and gateway soft probe system to collect data related to the quality of terminal equipment. Further, through the analysis of the massive historical data, AI algorithms can learn the changing pattern of TV service quality data on time series and finally find out the major abnormality.

The main contributions of this paper include:

- 1) designing and implementing a relevant procedure to detect video freezing and blurred screen in set-top box, extracting alarm event of video freezing and blurred screen as key indicators of Internet TV service quality data since it more close to user experience than network parameters, using the Soft Probe platform to complete real-time collection of alarm data from millions of active set-top boxes across the country. From the experimental results, video freezing and blurred screen alarm data is the very effective indicator of Internet TV service quality.
- 2) Adopting the Informer model for the detection algorithm for Internet TV service quality. Through comparison experiments, the Informer model performs better than other anomaly detection models in terms of detection accuracy and precision for such long-period time series of feature data and can be well applied to engineering practice.

II. RELATED WORK

Up to now, quite a few algorithms and models have been proposed on the task of time series anomaly detection. They

can be classified in a variety of ways, such as univariate and multivariate, supervised and unsupervised, or traditional and deep learning methods.

1) *Traditional methods*: Statistics-based anomaly detection models have the advantage of fast evaluation, but are usually only suitable for univariate data and are susceptible to noise. ARIMA [1] is a statistics-based model derived from auto-regression and moving average, it is widely used in time series research and has a series of variants, but it cannot capture unstable or nonlinear features, so it is rarely used for high-order multivariate time series anomaly detection. Among the distance-based anomaly detection model, a classic example is the KNN algorithm [2]. The principle is to calculate the anomaly score according to the distance between the sample point and its k-nearest neighbors. However, KNN is a lazy learning algorithm, which is expensive to evaluate, thus it is not suitable for scenarios with a large number of samples. Another example is the LOF algorithm [3], which judges whether the point is an anomaly point by comparing the density of each point and its neighborhood points, and the low density means the high probability of anomaly. In addition to the above two types of models, there are many other machine learning algorithms that perform well. The isolation forest algorithm [4], [5] regards outliers as anomalies. When we divide the data space with a hyperplane, the outliers with sparse distribution are more likely to fall into the subspace alone. The algorithm recursively constructs a binary tree in the same manner of constructing a decision tree, and selects the leaf node that is closer to the root node as the detection result to return. OCSVM [6], [7] makes improvements on the basis of SVM, maps data to high-dimensional feature space through kernel function, and constructs an optimal hyperplane. Different from SVM for binary classification tasks, OCSVM gets only one class. All points that do not belong to this class are detected as anomalies. The k-means algorithm [8] obtains k clusters by iteratively adjusting k cluster centroids, points far from the centroids are regarded as outliers. But for the time series anomaly detection task, the k-means algorithm only focuses on the data itself and not on the correlation, so it performs poorly.

2) *Deep learning methods*: With the increase in data size and data complexity, it is more difficult for traditional methods to cope with today's industrial and research scenarios. Deep learning has a strong potential to learn and deal with complex structures and causal relationships, so it is gradually showing its advantages in time series anomaly detection tasks. We roughly divide the models into prediction-based and reconstruction-based anomaly detection methods as follows.

The prediction-based anomaly detection models learn the historical trend of the data to predict the value of the current timestamp. They use the difference (prediction error) between the predicted value and the ground truth as the basis for anomaly detection. LSTM-NDT [9] uses LSTM [10] to predict the value of the next timestamp and NDT to generate an anomaly threshold. If the prediction error is higher than the threshold, it is judged as anomaly. DAGMM consists

of a compression network and an estimation network. The compression network trains a deep auto-encoder while the estimation network trains the parameters of GMM. DAGMM predicts the anomalies with the probability value output by GMM.

The reconstruction-based anomaly detection models encode the current data and decode the latent variables. It uses the difference (reconstruction error) between the reconstructed value output by the decoder and the original data as the basis for anomaly detection. MAD-GAN [11] implements the generator and discriminator in GAN [12] with LSTM. It uses the reconstruction error of the generator and the discrimination error of the discriminator to detect anomalies. Omnianomaly captures complex temporal correlations in sample space through GRU and uses VAE [13] to map observations to random variables. The author proposed a linear Gaussian state space model to concatenate random variables and GRU hidden variables. A planar NF technique is also employed to capture complex data distributions.

In addition, there are some models that use a combination of prediction error and reconstruction error, such as MTAD-GAT [14]. MTAD-GAT captures high-dimensional features using a 1-D convolutional layer and captures time-series features and temporal correlations using two parallel graph attention layers, followed by GRU to deal with time series patterns.

Informer is a Transformer based model. It is prediction-based when used for anomaly detection tasks.

III. DETECTION OF TV SERVICE QUALITY ANOMALIES

1) *Feature value selection and collection*: Usually, the quality of Internet TV service is directly related to the performance of CDN servers, transmission network, playback devices and so on. China mobile soft probe system can collect the relevant performance indicators of all TV users' set-top boxes and gateways, including network bandwidth, uplink rate, downlink rate, network jitters, weak light condition of gateways, as well as the video freezing, blurred screen, network transmission rate, network access method, video first frame loading rate, etc. In the process of feature selection, it is hoped that the feature values can fully match the actual experience of the users. From the perspective of users, the video freezing and blurred screen of the set-top boxes are very typical data to reflect the quality of internet TV services, while the network transmission rate, network jitters and other indicators can indirectly reflect the quality of TV services, however due to factors such as video encoding and resolution, it is difficult to accurately determine changes in user experience based on variations in network rate or network jitters. For example, when the user's network bandwidth deteriorates during peak periods, the playback server may adjust the resolution or frame rate to ensure the user's experience basically unchanged. Therefore, in the selection of feature values, we believe that the video freezing and blurred screen within the set-top boxes is more suitable as performance indicators to directly reflect the quality of Internet TV services.

The following is a graph of the daily change in the number of video freezing/blurred screen alarm users in province K where there are no large-scale TV service quality problems. From the figure, it can be seen that the number of freezing/blurred alarm users is not zero under normal circumstances but fluctuates within a certain range, and the fluctuation pattern is closely related to the changing pattern of the number of set-top box active users. This is due to the fact that set-top box users may experience certain personal issues, such as signal failure in the home WIFI network, failure of the set-top box itself, etc. The changing pattern of the number of active subscribers of set-top boxes is related to the daily viewing habits of users, which are usually busier at noon and in the evening, and also fluctuate periodically on weekends, in addition there are also temporary fluctuations due to the increase in the number of subscribers on holidays.

It is inevitable that a certain percentage of set-top box subscribers have individual faults, but operators are more concerned about the mass faults that lead to mass TV service degradation. Thus, quick detection and resolution of mass faults is the key to improving the quality of TV services. The following is a graph of the number of users in Province K that had a mass TV service quality problem. It can be seen that freezing/blurred alarm users increased significantly between 1am and 3am compared to the daily value in Fig. 2, which indicates that there was a serious mass fault event.

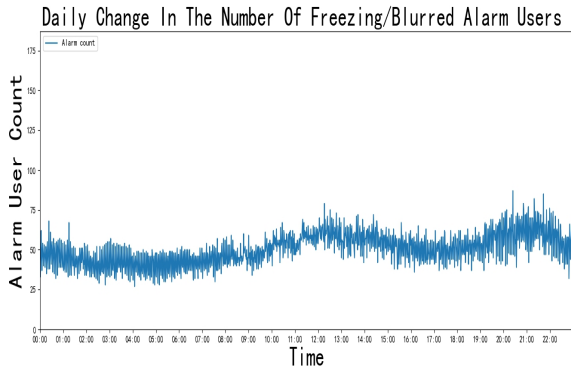


Fig. 2. Normal daily change in the number of alarm users

From the Fig. 3, it can be obtained that the number of historical freezing/blurred alarm users collected by the soft probe platform contains a large amount of user behavior information and hidden fault information. What the algorithm needs to do is to learn the information of these changing user behavior patterns and accurately identify the hidden faults.

A. Anomaly detection model

As mentioned above, the features input into the model is a set of long-term sequence values of the number of users per minute of video freezing or blurred screen warning reported by the soft probe platform for Internet TV services. It is expected that the model can effectively and accurately detect whether there are any anomalies for the feature set.

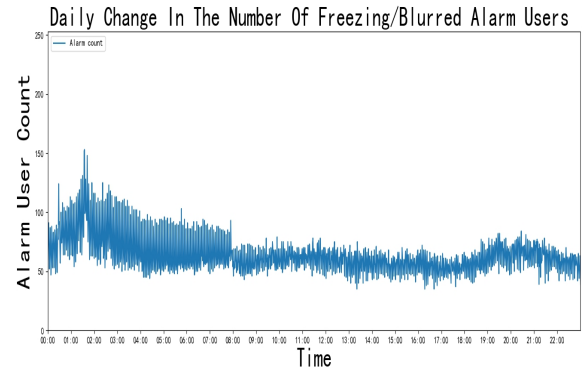


Fig. 3. Abnormal daily change in the number of alarm users

The current anomaly detection methods are mainly divided into prediction-based anomaly detection methods and reconstruction-based anomaly detection methods. For the prediction-based methods, it is a model that fits normal data by training on it. When anomaly occurs, since the model can only fit normal data, there is a difference between the predicted value and the real data. By judging the size of the difference, the location of the anomaly can be found. For the reconstruction-based methods, we mainly use AutoEncoder (AE) [15], Variational AutoEncoder (VAE) and Generative Adversarial Network (GAN) [12] and its derived deep learning models. By encoding the original input as a latent representation and reconstructing the data from the latent representation, it learns the patterns and reconstruction of normal data to determine whether it is an anomaly.

Among the prediction-based models, the Informer [16] has higher accuracy for long sequence time-series forecasting and is better than the Transformer in efficiency, which is more in line with the real-time requirements of the project. Therefore, this study selects the Informer as the prediction model. Further by comparing the actual effects of the prediction-based model Informer and the reconstruction-based model VAE, the result shows the Informer is more effective in detecting the features of long-period time series, so finally we adopted Informer for the real network deployment.

Since Informer is proposed as a prediction model and cannot be directly applied to anomaly detection of TV service quality, our study optimizes Informer to accommodate to the home internet TV service anomaly detection: The predicted value is compared with the actual value, and the large difference is judged as anomaly. Because the prediction part is an interval, in order to achieve the purpose of point-by-point detection and maximization of differences, only the last point of the prediction part is used to compare with the last point of the real part each time. As a result, point-by-point zero-latency anomaly detection can be achieved with the help of a sliding window. A threshold needs to be set when the method is applied, and it is judged as anomaly when the error between the predicted value and the actual value is greater than the threshold. The best way to determine the threshold is to train

and test on the non-abnormal training set, and choose the maximum error as the threshold. On the abnormal test set, any point with an error greater than the threshold is judged as an anomaly.

IV. COMPARISON OF TV SERVICE QUALITY ANOMALY DETECTION MODELS

The effect verification of the detection model is to compare the predicted values with the real values. The prediction values refer to the anomaly detection result of the model, and the label data is whether there are anomalies confirmed by operation and maintenance engineers through traditional empirical methods with various kinds of data and charts. The anomaly information obtained from the traditional empirical method is called an annotation during training.

The original input length, label sequence length, and predicted sequence length of the Informer are 96, 48, and 24 respectively. Considering the periodicity of the internet TV service time series, the cold start time and the time-consuming of reading data during deployment on the current network, the parameters adopted in this study are 60, 30, and 15, the input and output dimensions are both 1, which means the detection model is single-dimensional. There are 3 stacks with 3, 2, and 1 layer respectively in the encoder part and 2 layers in the decoder part. In addition to the information of year, month, week, day, hour and minute, holiday information is also considered when the timestamp is time-series encoded. The model is trained for 8 epochs with the learning rate of 0.0001 and the batch size of 32.

The anomaly detection results of the Informer for the TV service in province A are shown in the Fig. 4 (The time range is from September 26, 2021 to October 31, 2021). The green points indicate AI detection anomalies, and the red points indicate actual anomalies, the purple line indicates hypothetical alarm thresholds based on expert experience).

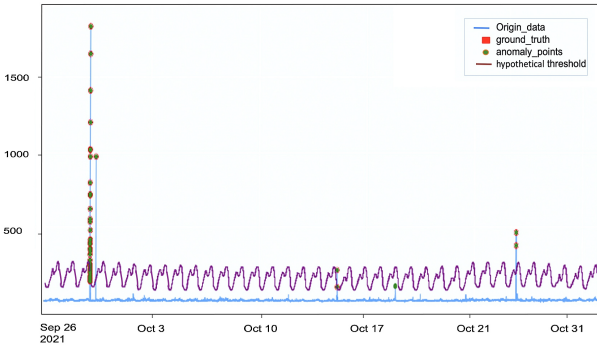


Fig. 4. First anomaly detection results for province A

The Fig. 5 shows the anomaly detection results of the Informer for province A from October 17th to December 4th.

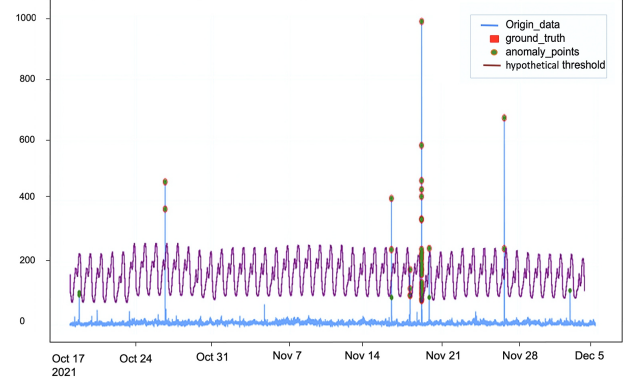


Fig. 5. Second anomaly detection results for province A

According to the Fig. 5 of province A, the accuracy, precision and recall are calculated as following TABLE I.

TABLE I
MODEL EVALUATION RESULT WITH PROVINCE A

Accuracy	Precision	Recall	TN	TP	FP	FN
0.999	0.927	1.000	100690	102	8	0

The Fig. 6 shows the anomaly detection results for province AC from September 26th to October 31th.

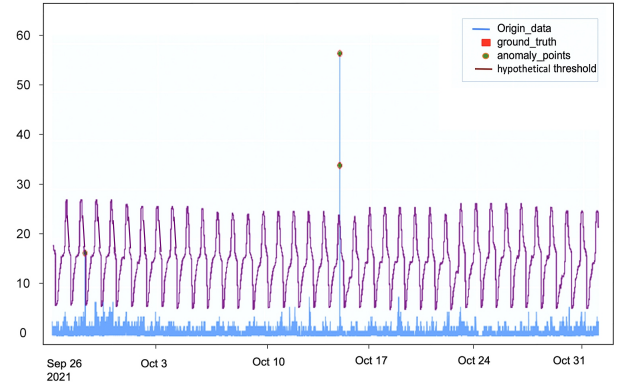


Fig. 6. First anomaly detection results for province AC

The Fig. 7 shows the anomaly detection results for province AC from October 17th to December 4th.

According to the above mentioned figures of province AC, the accuracy, precision and recall are calculated as following TABLE II.

As the comparison in this study, VAE is also a good anomaly detection model. It requires fewer main parameters, and the

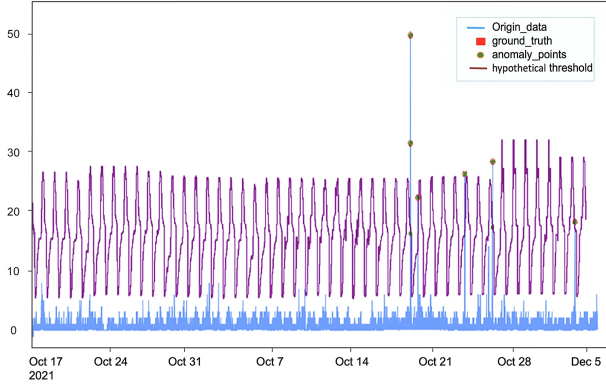


Fig. 7. Second anomaly detection results for province AC

TABLE II
MODEL EVALUATION RESULT WITH PROVINCE AC

Accuracy	Precision	Recall	TN	TP	FP	FN
0.999	0.818	1.000	100789	9	2	0

most important one of them is the window size of the input, which directly affects the accuracy and timeliness of model detection. The following figure is the detection result for the same data with the window size set to 60 and 5 respectively (green points for AI detection anomalies and red points for actual anomalies).

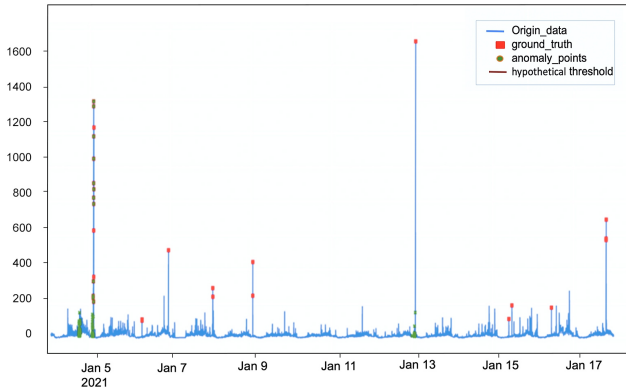


Fig. 8. VAE-v1 anomaly detection results with the window size of 60

From Fig. 8 and Fig. 9, it can be seen that VAE's overall detection effect performs well, it can capture almost all anomalies of the current network scheme. The detection effect of setting the window parameter to 5 is more accurate than setting it to 60. However, the figures above also show that VAE model is too sensitive for some situations. In general, the anomalies detected by the VAE almost cover the set of actual anomalies completely, so the recall rate of the model is high, but because

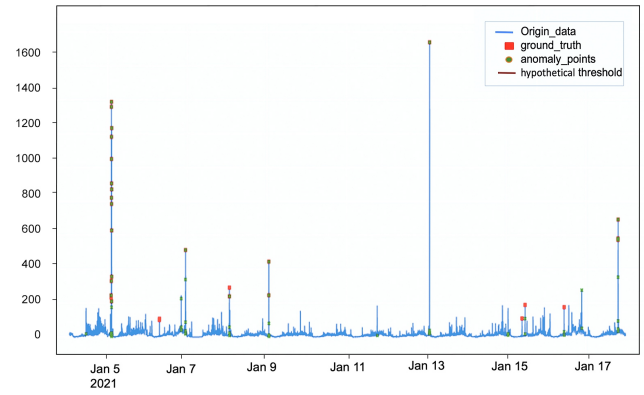


Fig. 9. VAE-v2 anomaly detection results with the window size of 5

of the model's sensitivity, some data points are misjudged as anomalies, which resulting in a low accuracy. The test effects of VAE are as following TABLE III.

TABLE III
MODEL EVALUATION RESULTS OF VAE

Model	Accuracy	Precision	Recall
VAE-v1	0.998	0.444	0.892
VAE-v2	0.999	0.596	0.994

TABLE IV shows the comparative operation effect of various machine learning and deep learning models on 31 provinces:

TABLE IV
MODEL RESULTS OF VARIOUS METHODS

Model	Accuracy	Precision	Recall	F1 Score
DIFF	0.9986	0.5000	0.999	0.666
KNN	0.9987	0.549	0.998	0.708
PCA	0.9988	0.551	0.999	0.710
OC-SVM	0.9987	0.548	0.997	0.707
IForest	0.9987	0.5185	0.998	0.682
SO-GOAL	0.7016	0.0048	0.999	0.009
AE	0.9988	0.5385	0.996	0.699
Integrated Model	0.999	0.5714	0.999	0.7269
VAE-v1	0.998	0.444	0.892	0.5928
VAE-v2	0.999	0.596	0.994	0.7451
Anomaly Informer	0.9991	0.8110	0.9139	0.8594

V. CONCLUSION

Based on the analysis of the above research, for the detection of abnormalities in home broadband TV service quality, using the number of freezing/blurred alarms on set-top box as indicator feature values, the Informer model can successfully detect serious TV service quality deterioration in real time, and the precision rate of anomaly detection reaches 80%, and the recall rate and accuracy rate reaches over 90%.

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